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A Multimodal Framework for Fatigue Driving Detection via Feature Fusion of Vision and Tactile Information

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Abstract

Driver fatigue is a major cause to traffic accidents, significantly impairing attention and reaction time. Traditional detection methods typically rely either on visual data or sensor signals. Image-based approaches suffer from lighting variations, while sensor-based methods are prone to noise interference. Here, a multimodal fusion architecture that integrates visual imagery with tactile signals from flexible sensors using porous composites is proposed to detect driver fatigue states. A convolutional neural network extracts features from the images while sensor signals are encoded through fully connected layers. The extracted representations are then projected into the same dimensional space for concatenated feature fusion. Experimental results show that the proposed multimodal approach improves recognition accuracy by more than 4% compared with single-modality methods and generalizes reliably across varied appearances and extreme conditions. Furthermore, a detailed evaluation and quantification of fatigue levels have been conducted, which contributes to accident prevention and promotes driving safety.

Kunpeng Li and Wei Yue contributed equally to this work.

Introduction

Driver fatigue, primarily caused by prolonged driving or insufficient sleep, severely impairs the driver's cognitive functions, including delayed reaction times, reduced attention, and increased instances of microsleep¹. Fatigue-related accidents account for over 20% of traffic incidents each year, underscoring the urgent necessity for effective prevention². Traditional regulatory measures, such as post-incident interventions and random inspections, fail to provide real-time, objective monitoring. This phenomenon has captured the attention of researchers, enabling potential solutions to combat driver fatigue, especially combining recent advancements in sensor technology, computer vision, artificial intelligence, and the Internet of Things (IoT)³⁻⁶.

Real-time fatigue detection methods can be classified into two categories based on the data source: behavior-based approaches and physiology-based signal analysis. Behavior-based methods primarily use computer vision technology to monitor facial features such as eye openness, blink frequency, and yawning to assess fatigue levels⁷⁻¹⁰. Li et al. developed a fatigue detection algorithm using facial multi-feature integration, employing an improved YOLOv3-tiny network with supported vector machine (SVM) classifiers to analyze eye and mouth states, and achieved real-time detection with 95.1% accuracy¹¹. However, this approach is highly susceptible to external factors such as lighting variations, facial occlusions (e.g., glasses or masks), and individual differences, which significantly reduce the detection accuracy. Physiological signal-based methods assess driver fatigue by collecting data such as electroencephalography (EEG), electromyography (EMG), and electrooculography (EOG). These signals provide a direct and real-time reflection of a driver's state with rich information while being less affected by external conditions and individual variability¹²⁻¹⁵. Zheng et al. designed a wearable EOG device with flexible dry electrodes for real-world vigilance estimation and successfully evaluated driver fatigue using temporal dependency models¹⁶. This method exhibits relatively low susceptibility to environmental factors such as light and temperature. However, it faces certain limitations: firstly, the majority of existing electrodes are rigid, resulting in discomfort during prolonged use, and secondly, the sensors are often bulky, further restricting user mobility¹⁷⁻¹⁹. Additionally, the inherent complexity of signal characteristics presents considerable challenges for practical implementation, requiring accurate extraction and algorithm assistance (Supplementary Table 1)

Therefore, given the performance degradation of vision systems in low-light conditions and the noise susceptibility of wearable devices, multimodal fusion has emerged to enhance perceptual accuracy and system robustness, garnering significant interest²³⁻²⁶. Integrating vision data with sensor signals enables comprehensive and real-time assessment of a driver's state. Beyond improving detection precision, multimodal fusion enhances robustness under challenging conditions, including low-light environments, camera occlusion, and partial data loss, ultimately improving driving safety and user experience²⁷⁻²⁹. The primary challenge in multimodal fusion is the complexity of integrating data across multiple dimensions and varying sampling frequencies^{30,31}. Vision data consists of continuous two-dimensional (2D) or three-dimensional (3D) image frames, whereas physiological and force sensor signals are typically one-dimensional time series. Effective fusion strategies for these heterogeneous data sources are essential to ensure the accuracy of both vision and physiological data in fatigue detection.

To tackle this challenge, this study proposed a convolutional neural network (CNN)-based framework for vision and sensor fusion, aimed at achieving high-precision driver fatigue detection (Fig. 1a). Particularly considering the human brain's somatosensory-vision interaction mechanism, which relies on dynamic interactions among regions such as the posterior parietal cortex and superior temporal areas. In these areas, signals from tactile and vision patterns converge to ensure coherent perception and guide motor actions³². Consequently, the sensor configuration is strategically placed, as depicted in Fig. 1b. To enable real-time physiological monitoring, flexible sensors were developed using a porous composite of poly styrene-block-butadiene-block-styrene (SBS) and carpenterworm-like polyaniline (C-PANI) (Figs. 1c, d). The data from these sensors and vision data captured under varying lighting conditions were used to construct a multimodal driver dataset. A sliding window approach was used to collect sensor signals, while image acquisition frequency was synchronized with the signal to enable real-time detection. A 2D-CNN processes vision information in this framework, while a 1D-CNN extracts tactile features, mapping them into a common feature space for deep feature fusion. The proposed method effectively identifies fatigue-related behaviors such as closing eyes, head nodding, and yawning, achieving an average classification accuracy of 98.2% under standardized testing conditions. We evaluated generalizability across varied appearances and extreme conditions, where performance declined but accuracy remained reasonably high, demonstrating the robustness of the multimodal fusion architecture. Additionally, by coupling model predictions with a predefined rule set, we defined

fatigue levels spanning mild to severe, enabling a graded assessment–intervention framework that underscores the joint hardware–algorithm potential of our approach for practical driver fatigue monitoring.

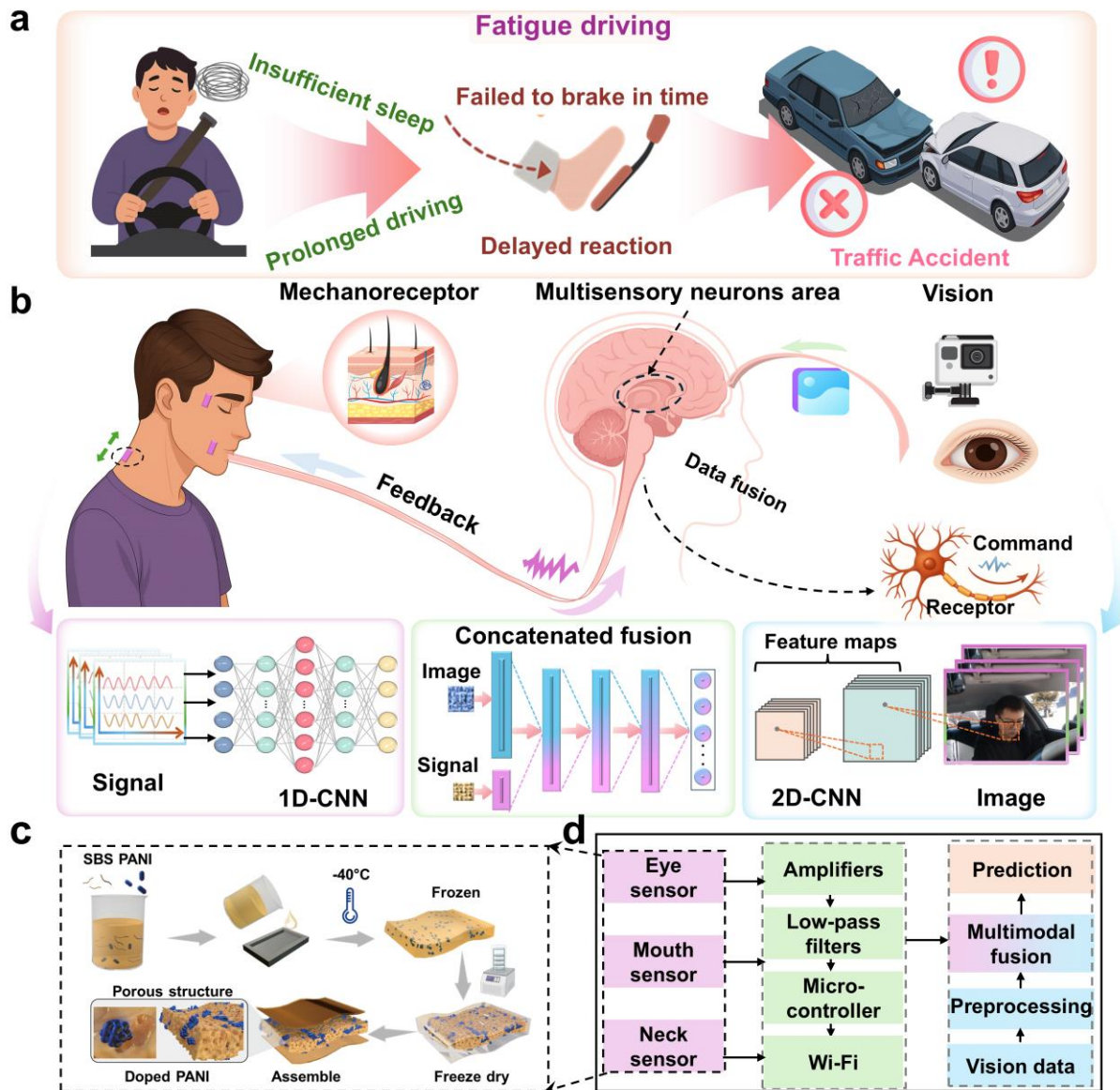


Fig. 1 Illustration of multimodal feature fusion for fatigue driving detection. (a) Car accidents caused by driver fatigue. **(b)** Schematic of vision and tactile information processing in the human brain and framework for sensor-vision data fusion using CNN. **(c)** Schematic illustration of the fabrication of flexible sensors composed of porous composites. **(d)** System-level diagram of multimodal driver fatigue monitoring.

Results

Design and Sensing Performance of the Sensor

To fulfill the requirements for high sensitivity and lightweight sensors suitable for long-term wear, the choice of the material system is crucial. Porous composites have garnered significant attention in recent years due to their promising applications in flexible sensing^{33,34}. These materials are typically fabricated by dispersing conductive fillers into a polymer matrix, followed by a foaming process. The resulting porous structure imparts several advantages, including lightweight, compressibility, and ease of processing^{35,36}. SBS is a cost-effective thermoplastic elastomer known for its excellent mechanical properties and low-temperature resistance, facilitating the development of highly elastic and durable porous composites^{37,38}. Incorporating a porous structure into SBS-based composites reduces the weight of their framework with enhanced specific surface area and porosity, leading to superior mechanical flexibility, compressibility, and sensing responsiveness³⁹. Incorporating conductive polymers into the system can enhance both conductivity and sensitivity, where PANI stands out for its low cost, ease of synthesis and high electrical conductivity^{40,41}. PANI can be fabricated into a variety of well-controlled morphological and dimensional nanostructures with corresponding performance, such as nanofibers, nanotubes, nanowire arrays, and even 3D hierarchical micro/nanostructures (e.g., coral-like, flower-like, and sea urchin-like structures)⁴². These structures offer a large specific surface area and optimized ion diffusion pathways, further enhancing sensing capabilities^{43,44}.

Using SBS and PANI as primary components of the system, the detailed preparation process of the porous C-PANI/SBS composite is described in the Experimental Section. As shown in Fig. 1c, assembling the synthesized composite with Cu/PI thin electrodes results in a flexible tactile sensor. To confirm the successful fabrication of C-PANI/SBS foam, scanning electron microscopy (SEM) was used to characterize its microstructure at different magnifications. The four SEM images in Fig. 2a depict an increasing magnification sequence, revealing micron-sized porous structures and the successful incorporation of C-PANI into the SBS foam, forming a porous composite. The last two images display SEM images of C-PANI, which exhibit a 3D carpenterworm-like hierarchical nanostructure. C-PANI consists of coarse nanofibers with an average diameter of approximately 120 nm, covered with short and fine nanorods. Furthermore, a computational model of the C-PANI/SBS foam was developed, and finite element simulations were performed under different pressure conditions (Fig. 2b). The results indicate that the porous structure effectively enhances pressure sensing due to its high porosity and low compressive modulus. Moreover, under high compression, for the same compression distance and electrode area, C-PANI/SBS foam benefiting

from the addition of SBS exhibits a higher dielectric constant compared to pure SBS foam, enabling greater pressure sensitivity and a broader response range.

The sensing performance of the sensor was evaluated by analyzing C-PANI/SBS foam relative capacitance changes and sensitivity (Fig. 2c). Sensors composed of a solid SBS layer, pure SBS foam, and SBS foam with C-PANI were compared under applied pressures of up to 300 kPa. Among them, the C-PANI/SBS-based sensor demonstrated the highest sensitivity, reaching 0.98 kPa^{-1} in the low-pressure range and 0.06 kPa^{-1} in the high-pressure range, with a slower response saturation as pressure increased. A further assessment of the resolution of the porous C-PANI/SBS composite sensor under high pressure showed that the response to a slight pressure of 2.3 kPa remained distinguishable even when 80 kPa was applied, exhibiting step-like patterns and superior resolution (Fig. 2d). As shown in Fig. 2e, the sensor was validated at its minimum detection limit (5.3 Pa), showing a capacitive response that consistently exceeded ambient noise in repeated applications. The sensor exhibited response and recovery times of less than 20 ms under external pressure loading and unloading (Fig. 2f). Additionally, the sensor was subjected to loading and unloading sequences at different pressure levels to examine its dynamic response. Fig. 2g shows the response patterns of the sensor, which were similar when subjected to 6, 23, 70, 120, and 188 kPa pressures, repeated three times. The capacitive response remained stable during the loading phase and returned to its original state after pressure removal, indicating excellent repeatability and recovery. Fig. 2h shows that during the dynamic process of continuously increasing and decreasing the bending angle of the sensor, the capacitive signal response consistently maintains excellent repeatability. The durability and stability of the sensor were evaluated by repeatedly applying a pressure of 68 kPa. The sensor exhibited highly stable responses after over 20,000 loading and unloading cycles (Supplementary Fig. 1). The inset of Supplementary Fig. 1 illustrates the capacitive response over five cycles at the beginning (left) and near the end (right) of a pressure stability test. Additionally, Supplementary Fig. 2 shows the impact of cyclic bending on the sensor, investigated by subjecting it to a 70° bending angle for over 15,000 cycles. No significant drift in the sensor's response was observed, demonstrating the developed sensor's excellent stability and reliability.

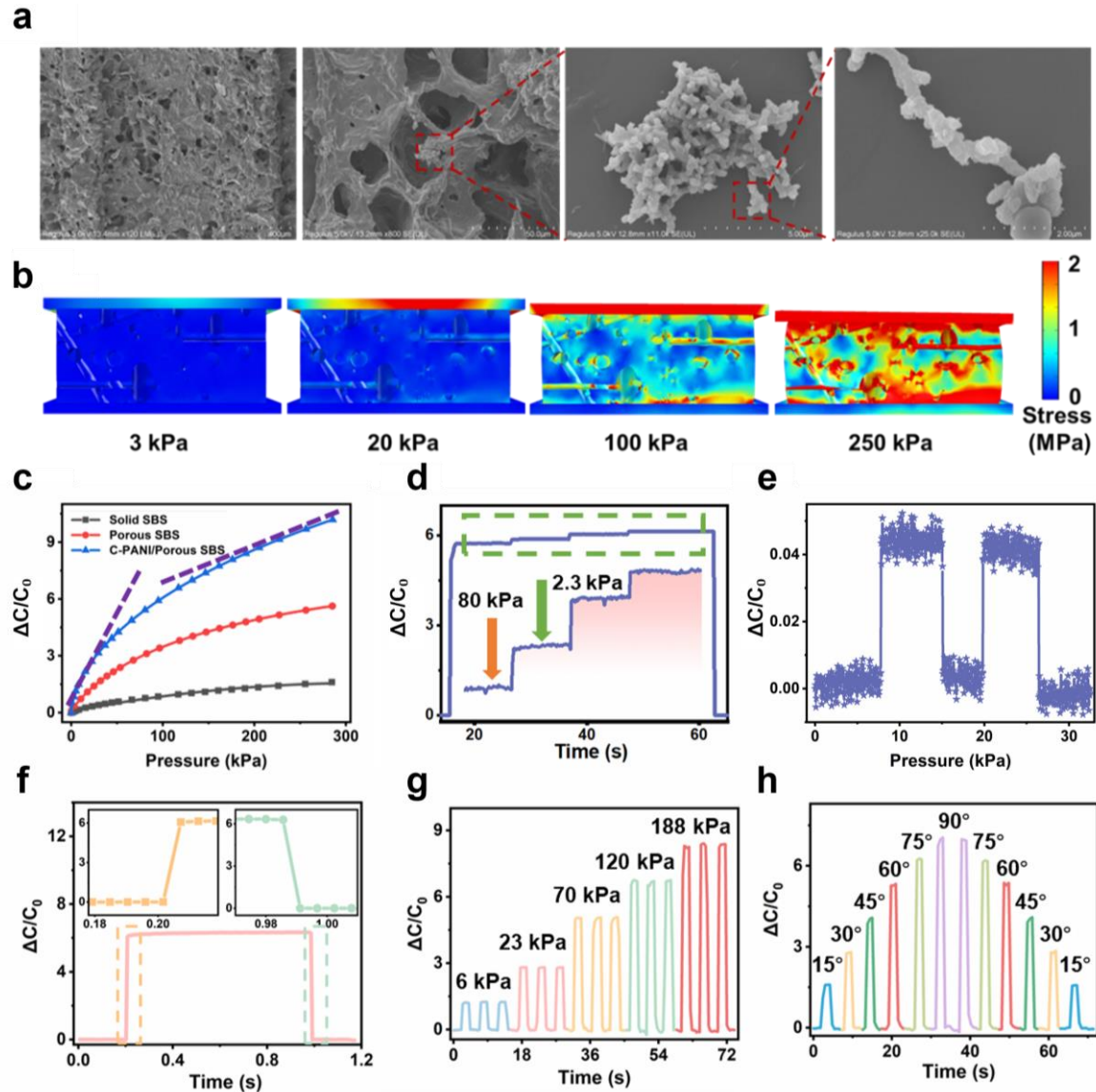


Fig. 2 (a) SEM images of the dielectric layer characterizing micron-scale pores and nanoscale C-PANI at different magnifications. (b) Finite element simulation results of the cross-sectional stress distribution in the SBS foam structure model under different pressures. (c) Relative capacitance changes of three capacitive tactile sensors: solid SBS layer, pure SBS foam, and SBS foam with C-PANI. (d) Relative capacitance changes of the sensor for slight pressure variations under high pressure conditions. (e) Consecutive capacitive responses at the limit of detection. (f) Response and recovery times of the sensor. (g) Capacitive response of the sensor to repeatedly applied gradient pressure. (h) Capacitive response of the sensor to continuous gradient increases and decreases in the bending angle.

Construction of Vision and Sensor Dataset

To construct an effective vision and sensor dataset, it is essential to first establish a mapping relationship between sensor data and indicators of driver fatigue. Driver fatigue is typically characterized by observable behaviors, such as frequent closing eyes, yawning, and repetitive nodding. Our fusion framework strategically integrates tactile sensors positioned in three critical regions — the eyes, mouth, and neck — complemented by vision capture devices oriented toward these areas, enabling continuous monitoring and data collection for comprehensive assessment (Fig. 3a). Detailed, the mouth sensor captures movement variations associated with yawning, while the eye sensor, positioned near the orbital region, records eyelid closure. The neck sensor, attached to the posterior region, measures pressure changes related to nodding. Additionally, a commercial wireless camera was positioned above the dashboard to unobtrusively capture variations across these three regions. These two types of signals are gathered as the raw signals constituting the vision and sensor (VS) dataset.

Further, the raw vision and tactile signals undergo the data processing workflow illustrated in Fig. 3b to achieve normalization and noise reduction. Specifically, capacitance-based signals from the tactile sensors are subjected to low-pass filtering and normalization to mitigate hysteresis effects, while the RGB images captured by the camera are resized for uniform dimensions and augmented to enhance data diversity. The processed signals are referred to the validated VS dataset and are subsequently partitioned into three distinct subsets: training (70%), validation (10%), and testing (20%) subsets to facilitate model development and performance evaluation. Building upon the aforementioned data collection and processing workflow, the driver's state in a driving environment was simulated to generate a dataset reflective of real-world scenarios. As illustrated in Fig. 3c, four distinct driver states were simulated under two different lighting conditions. The “Normal” state represents an alert and attentive driver, while the remaining states correspond to typical manifestations of driver fatigue, including closing eyes, nodding, and yawning, respectively. Vision image signals and capacitance signals from three tactile sensors collected from these states serve as the data sources for the VS dataset. The effectiveness of vision images in recognizing fatigue behaviors was distinctly evident, but images are uninformative in low-light conditions. Then, the effectiveness of capacitive sensor data in accurately identifying fatigue driving behaviors was evaluated. As illustrated in Fig. 3d, the capacitive response curves for three distinct channels—eyes, neck, and mouth—are presented alongside their corresponding heat maps, showing the sensor responses under three typical fatigue behaviors: closing eyes, nodding, and

yawning. Obviously, the eye sensor (orange curve) exhibits a significant drop during closing eyes, remaining stable in other states. However, the neck sensor (blue curve) exhibited a pronounced decrease during head nodding, with substantial movements inducing concomitant fluctuations in the other sensors. The mouth sensor (red curve) displayed a clear periodic response during yawning, and the pressure exerted by yawning correspondingly influenced the other sensors. Additionally, due to dimensional differences between signal data and images, directly analyzing their correlation often lacks intuitiveness. To address this, we adopted a heatmap technique that maps one-dimensional signal data into a two-dimensional matrix (constructed from signals with dimensions of 3×16), using color intensity variations to depict capacitance values fluctuations visually. For instance, when the heatmap exhibits significant color changes, it indicates that the sensor has detected a potential fatigue state in the driver. This method significantly improved the readability of the data.

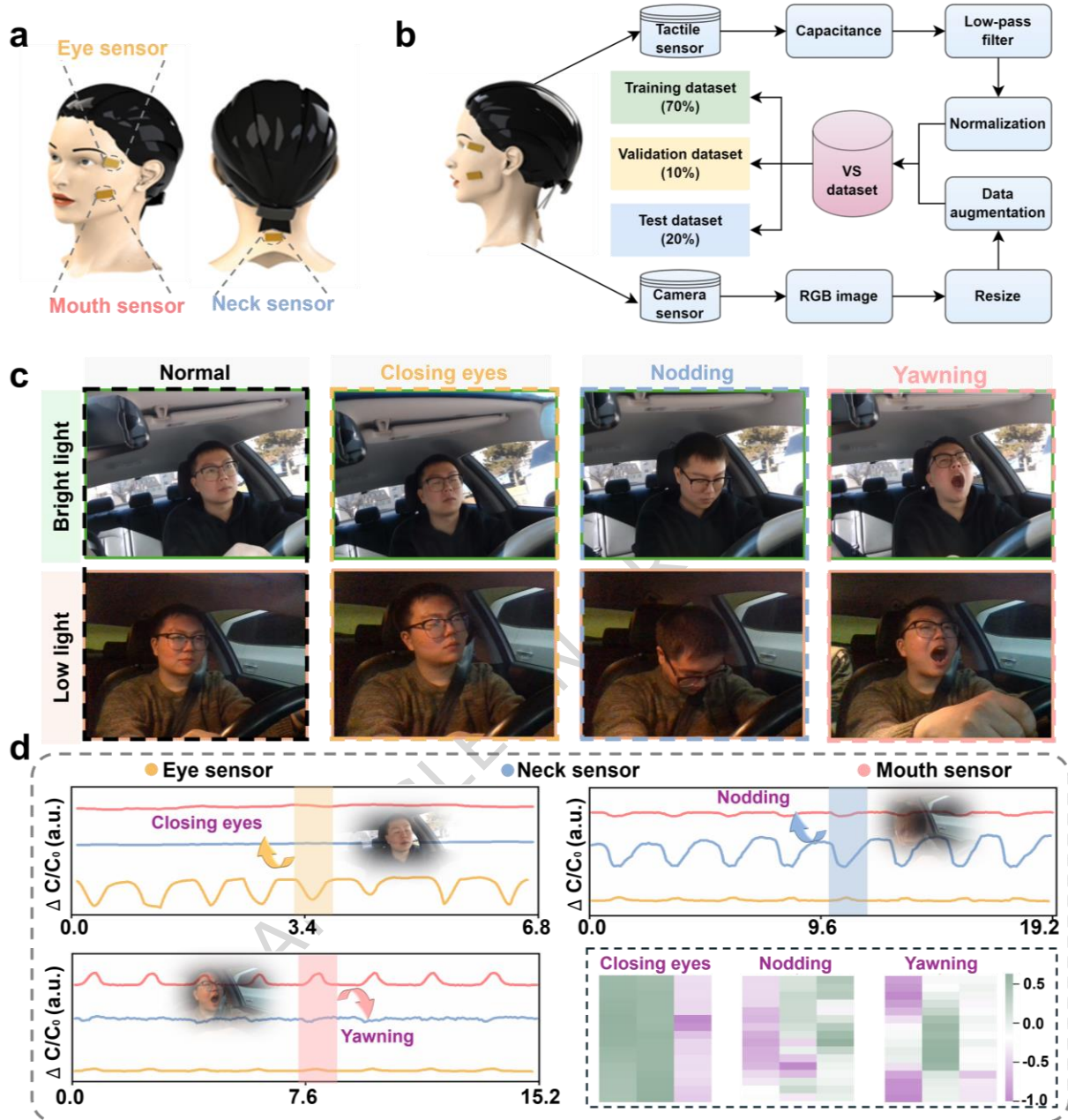


Fig. 3 Data preparation for fatigue driving detection based on vision and sensor fusion. (a) Placement illustration of a flexible sensor on the mouth, eye, and neck. (b) Flow diagram for data collection to construct the VS dataset from tactile and camera sensors. (c) Illustration of four driving states under different illuminance conditions with vision images. (d) Real-time three-channel waveforms and corresponding heatmaps for three fatigue states.

Construction of the Multimodal Feature Fusion Model

Based on the constructed VS dataset and the validated effectiveness of unimodal signals (vision and tactile), a fusion strategy is implemented to enhance accuracy and address the limitations of

individual modalities. As shown in Fig. 4a, a deep learning-based multimodal architecture is proposed, which primarily consists of a vision feature extraction backbone and a data fusion module. EfficientNet serves as the backbone, leveraging a series of seven MBConv blocks to extract deep vision features from the RGB images ($3 \times 224 \times 224$), with detailed structural parameters provided in Supplementary Table 2⁴⁵. Transfer learning is employed to enhance training speed and accuracy, using pre-trained weights from the ImageNet dataset as the initial parameters^{46,47}. Simultaneously, tactile sensor data (3×16) is encoded through three fully connected layers before concatenating with the extracted vision features for feature fusion. The feature vector dimensions are 1280D and 128D for the vision and tactile sensor features, respectively. A unified feature vector is generated by fusing features through the concatenation method. This fused feature is then fed into a fully connected layer for further processing, and finally, the driver's fatigue behavior is classified and predicted through a Softmax function. The training loss curves depicted in Supplementary Fig. 3 indicate that the model successfully converges after 200 training epochs. The validation loss follows a similar downward trend, demonstrating good generalization performance without any signs of overfitting. The model effectively classifies four driving states: Closing eyes, nodding, normal, and yawning. To ensure temporal alignment between the vision and tactile streams, both modalities were recorded in the same acquisition program on a single host computer using a shared system clock, with the tactile stream (12.5 Hz, 16-sample sliding window, 80 ms update) as the temporal reference after a 3~5sec resting period to stabilize the baseline. Each tactile window was assigned to the time stamp of its last sample and matched to the video frame with the closest time stamp, so that the residual misalignment was bounded within one video-frame interval, which is smaller than the tactile update interval⁴⁸.

Fig. 4b presents the confusion matrix, demonstrating an average accuracy of 98.2%, highlighting the model's strong classification capability. To further evaluate its performance, t-distributed stochastic neighbor embedding (t-SNE) was used to visualize the test set data by projecting high-dimensional feature representations onto a 2D space, as shown in Fig. 4c⁴⁹. The resulting well-separated clusters indicate strong feature discriminability, with clear boundaries validating the ability of the multimodal fusion strategy to effectively distinguish different fatigue states. By integrating the vision and tactile-based signals, this framework significantly enhances the robustness and accuracy, ensuring reliable and adaptive driver monitoring across various

conditions. We utilized the gradient-weighted class activation mapping (Grad-CAM) method to visualize the feature maps at different layers during the image feature extraction process⁵⁰. Supplementary Fig. 4 shows that the feature distributions exhibit significant variations across various states, particularly in the eye, mouth, and neck regions. These variations highlight the model's ability to capture distinctive spatial patterns associated with each state. As a further comparison, we developed two unimodal methods for driver state detection: a vision-based recognition method using only images, implemented with a 2D-CNN (Supplementary Fig. 5), and a signal-based recognition method relying solely on tactile sensor data, implemented with fully connected layers (Supplementary Fig. 6). A receiver operating characteristic (ROC) curve was utilized to evaluate the predictive performance across different thresholds. Sensitivity and specificity for each threshold were calculated as defined in Equation (1) and (2), respectively.

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (1)$$

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \quad (2)$$

where TP represents correctly predicted positive instances, FN refers to actual positive instances incorrectly predicted as negative, TN represents correctly predicted negative instances, and FP refers to actual negative instances incorrectly predicted as positive. Figs. 4d–f illustrates the ROC curves of three different fatigue detection models: sensor-based (Supplementary Fig. 7), vision-based (Supplementary Fig. 8), and our approach. Compared to unimodal models, the VS model achieves the highest area under the ROC curve (AUC), demonstrating superior classification sensitivity and specificity. To more rigorously evaluate the benefit of fusion, we repeated the training–testing procedure five times with different random seeds. As shown in Fig. 4g, the vision-only and signal-only models achieved $92.9 \pm 0.7\%$ and $85.8 \pm 1.8\%$ accuracy, respectively, while score-level fusion baselines AF1 and AF2 yielded $94.1 \pm 0.7\%$ and $95.0 \pm 1.0\%$. Our feature-level AF attained $97.9 \pm 0.4\%$ accuracy, clearly outperforming both unimodal and late-fusion strategies and confirming that it better exploits complementary vision and tactile information. Additionally, we conducted an ablation on the fusion layer, including an attention-based variant. As shown in Supplementary Table 3, the three-layer fully connected (FC) fusion yields the best performance and slightly surpasses the attention-based model, indicating that the extra attention layer offers no clear benefit in our low-dimensional two-stream setting⁵¹.

We additionally conducted a comparison across alternative vision backbones under identical augmentation, optimizer, number of training epochs, and fusion head. As reported in Supplementary Table 4, EfficientNet achieves the highest accuracy while retaining the smallest parameter count⁵²⁻⁵⁴. To further validate the effectiveness of our method, we benchmarked it against the approaches of Du et al.⁵⁵, Shi et al.⁵⁶, Hassan et al.⁵⁷, Xing et al.⁵⁸, and Li et al.⁵⁹ on the VS dataset under a unified evaluation protocol. As shown in Supplementary Table 5, our results indicate that with modest training data, lightweight fusion combined with strong modality-specific encoders is less data-hungry and achieves better regularization compared to attention-heavy cross-modal Transformers, which typically require larger pretraining corpora or stronger regularization to reach their full potential. Notably, when visual data are degraded by illumination-induced noise, late fusion struggles to leverage the comparatively clean tactile signals to correct the corrupted visual cues. By contrast, although early feature fusion assumes strict temporal alignment and risks entangling modality-specific noise, concatenation after unimodal encoding preserves each modality's integrity and allows the classifier to exploit complementary cues without overmixing noise.

Subsequently, we conducted a comprehensive evaluation of the model's generalization performance in real-world scenarios, taking into account demographic diversity as well as challenging environmental conditions. We collected additional data from four cohorts representing demographic and appearance variability—older adults, individuals wearing medical masks, those with darker skin tones, and those with dense facial hair. Representative examples are shown in Supplementary Fig. 9, with quantitative results in Supplementary Table 6. The model maintained strong performance across these cohorts, indicating robustness to moderate variations in age and appearance. We further stress-tested the system under extreme conditions, including very low illumination, severe motion artifacts and degraded sensor–skin adhesion (Supplementary Fig. 10; Supplementary Table 7). Although performance declined under these stressors, accuracy consistently remained $\geq 85\%$, demonstrating the resilience of the proposed approach to challenging real-world scenarios and supporting its practical viability. Moreover, we evaluated a dynamically weighted fusion (DWF) mechanism under extreme conditions. As shown in Supplementary Table 7, its performance is comparable to our method, offering no substantial quantitative gain over our fusion strategy and confirming that our multimodal system already copes effectively with fluctuations in modality reliability. We further assessed robustness to temporal drift between the

tactile and image streams by shifting the signal features by k ($k=1,2$, corresponding to ± 80 ms, ± 160 ms, respectively) tactile steps and by randomly permuting the vision windows before fusion. The results are presented in Supplementary Table 8. For small shifts (± 1 step), accuracy decreased only marginally compared with perfect alignment. For larger shifts (± 2 steps), the drop was more evident but still above or comparable to the better unimodal baselines. In contrast, random pairing reduced multimodal performance to around or below unimodal levels, indicating that our concatenation-based late fusion tolerates mild misalignment but its benefits still rely on meaningful temporal alignment.

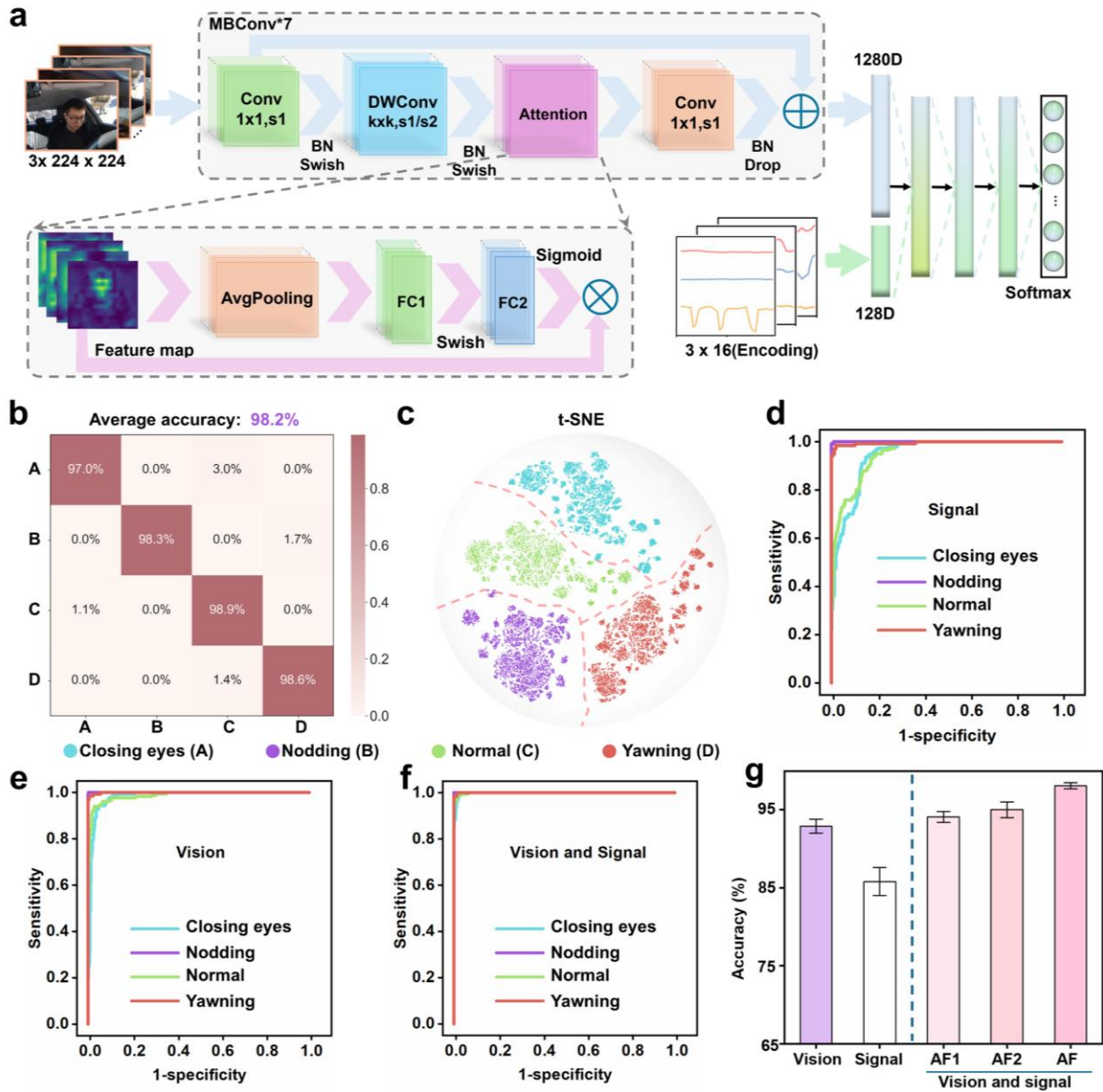


Fig. 4 Multimodal model for fatigue driving detection. (a) Detailed architecture of the multimodal model. (b) Confusion matrix for predicting four driving states with an average accuracy of 98.2%. (c) t-SNE visualization of feature distribution for the four driving states. (d) ROC curve of sensor-based fatigue detection. (e) ROC curve of vision-based fatigue detection. (f) ROC curve of VS fatigue detection. (g) Vision-based recognition uses only images ($92.9 \pm 0.7\%$), and signal-based recognition uses only tactile sensor data ($85.8 \pm 1.8\%$). AF1: weighted averaging fusion ($94.1 \pm 0.7\%$), AF2: weighted-multiplication fusion ($95.0 \pm 1.0\%$). AF: concatenation fusion ($97.9 \pm 0.4\%$).

Development of the Multimodal Fatigue Detection System

By integrating the proposed fusion strategy with embedded processing units, including Wi-Fi wireless communication modules, we developed an intelligent system capable of real-time monitoring of the driver's fatigue status. Fig. 5a illustrates the system architecture, where the sensor configuration remains consistent with the previously described setup. The acquired data is transmitted to the smart terminal via Wi-Fi for further processing. Given that fatigue manifests in varying degrees, each requiring different interventions, the monitoring system classifies fatigue levels into three categories: Normal, Mild Fatigue, and Severe Fatigue, based on specific indicators. To establish these thresholds, the normal physiological range of the human body was considered. For eye closing frequency, values below 18 are categorized as Level 1, values between 18 and 30 indicate Level 2, and values ≥ 30 signify Level 3. For nodding frequency, values < 2 are considered Level 1, 2–4 indicate Level 2, and ≥ 4 indicate Level 3. Similarly, values < 2 for yawning frequency are classified as Level 1, 2–4 as Level 2, and ≥ 4 as Level 3. All frequencies are calculated in times per minute. A comprehensive scoring system was defined to summarize the fatigue state: 0 points for Level 1, 1–2 points for Level 2, and ≥ 3 points for Level 3. A multimodal model was employed to extract deep features from VS data. Building on the ability of the multimodal framework to classify the three physiological states, the model further refines fatigue assessment by assigning one of three levels. Fig. 5b illustrates this workflow, beginning with acquiring face images and tactile signals, followed by a feature-fusion step and subsequent classification. As shown in the “Compute Metrics” stage in Fig. 5b, three key physiological indicators are counted and collected: closing eyes frequency, nodding frequency, and yawning frequency. Each metric is evaluated against specific thresholds to categorize the driver's condition. A final scoring scheme (0 points for Normal, 1–2 points for Mild Fatigue, and ≥ 3 points for Severe Fatigue) encapsulates these individual metrics into a single fatigue index, thereby enhancing both the accuracy and

interpretability of the model's driver-fatigue evaluation. To accommodate inter-individual differences in behavior, the system includes a dedicated calibration interface (Supplementary Fig. 11). During a brief self-reported alert period (e.g., 3–5 min), it estimates driver-specific baseline frequencies (mean and variance) of closing eyes, nodding, and yawning, and accordingly updates the detection thresholds to better match each driver's natural blinking and yawning rates.

The real-time fatigue detection results are shown in Fig. 5c (Supplementary Video 1). This system provides real-time vision feedback of the driver's fatigue along with tactile signal data, enabling the multimodal model to integrate features from multiple sensors and efficiently predict the driver's fatigue state. The system continuously computes and records fatigue indicators and triggers alerts when necessary to mitigate the potential risks associated with fatigued driving (Fig. 5d). For instance, when mild fatigue is detected, the system issues a warning, prompting the driver to take a break. In cases of severe fatigue, a mandatory alert is triggered, requiring the driver to stop driving immediately and rest to ensure road safety. Finally, to facilitate real-time in-vehicle monitoring, we deployed the model on an NVIDIA Jetson Xavier NX. As shown in Supplementary Fig. 12, the average power consumption was 7.42 W, and the model-side inference throughput reached 29.8 FPS (~ 33.6 ms/inference), thereby satisfying the power and latency requirements for continuous, real-time operation in vehicles.

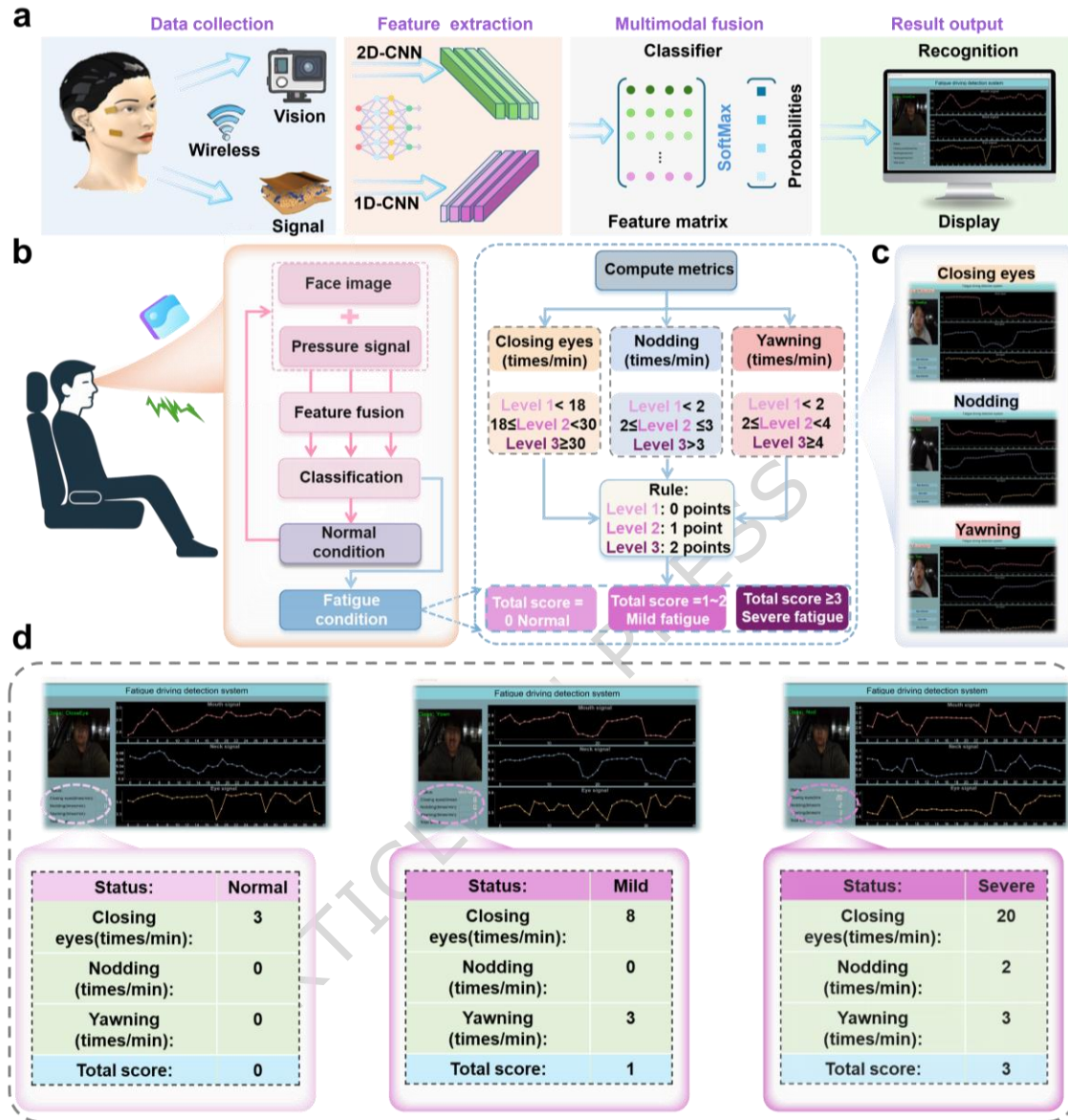


Fig. 5 Demonstration of fatigue driving detection in actual low-light environments. (a) Illustration of the process of multimodal data pre-processing, feature fusion, and prediction for fatigue driving monitoring. **(b)** Decision framework for multimodal fatigue monitoring based on vision and tactile signals. **(c)** Real-time prediction of fatigue status. **(d)** Detailed evaluation of fatigue status.

Discussion

We propose a multimodal fatigue detection method that integrates vision imagery and sensor signals to identify driver fatigue states accurately. The proposed system utilizes flexible tactile sensors fabricated from porous C-PANI/SBS composites with well-defined morphologies. These sensors can be conformally attached to the driver's skin, enabling the reliable detection of subtle tactile signals. Concurrently, a commercial wireless camera captures real-time RGB images, and

a sliding window strategy ensures synchronized acquisition of both sensor signals and image data. Compared to common fusion architectures, our concatenation-based approach demonstrates improved accuracy, even under low-light conditions. This fusion strategy effectively mitigates the sensitivity of traditional vision-based methods to brightness variations while also reducing instabilities introduced by signal noise. It also maintains reliable performance across diverse appearance attributes and under extreme conditions. Moreover, leveraging the model's three fatigue states, we designed a rule-based scoring scheme that translates frame-level predictions into a quantitative fatigue index and triggers timely driver alerts. We further deployed the model on an NVIDIA Jetson Xavier NX for on-device inference, achieving 29.8 FPS, which meets real-time requirements. Across diverse and challenging environments, the proposed multimodal fusion framework maintains stable, high-quality performance, underscoring its potential as a reliable solution for driver fatigue monitoring and early intervention. Future work will address the need for more diverse participants and real-world on-road validation by recruiting a broader cohort and conducting on-road field tests.

Methods

Materials

Aniline (analytical grade, >99.5%) and 1,4-dioxane (purity >99.0%) were purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd. Ammonium persulfate (APS), hydrochloric acid (HCl), and ethanol (analytical grade, >99.7%) were purchased from Sinopharm Chemical Reagent Co., Ltd. SBS granules ($M_w \sim 140000$) were sourced from Guofeng Rubber & Plastic Co., Ltd. The detailed specifications of all chemicals are provided in Supplementary Table 9. All chemicals were used as received, without further purification.

Fabrication of Carpenterworm-like Polyaniline (C-PANI)

The C-PANI micro/nanostructures were synthesized via a simple frozen interfacial polymerization method. First, aniline and APS were dissolved in a 1 M HCl aqueous solution. Then, 10 mL of the APS solution was frozen into ice layers in a 50 mL glass beaker at -20°C . Next, 10 mL of a pre-cooled aniline solution (0°C) was carefully poured onto the oxidant ice layer in the beaker. The mixture was subsequently frozen at -20°C for 2 h before being transferred to an ice bath for an additional 10-hour reaction. The resulting dark green product was filtered, thoroughly washed multiple times with deionized water and ethanol, and dried in a vacuum oven at 60°C overnight to obtain the C-PANI micro/nanostructures.

Fabrication of Porous C-PANI/SBS Composite

SBS granules were fully dissolved in 1,4-dioxane under mechanical stirring at 60°C for 4 h (Fig. 1c). The weight-to-volume ratio of SBS granules to 1,4-dioxane was set at 6:50 (g/mL). The resulting SBS/1,4-dioxane solution was poured into a polytetrafluoroethylene (PTFE) mold and left to evaporate at 30°C overnight, forming control samples of solid SBS layers. For porous C-PANI/SBS foam, C-PANI was dispersed in the SBS/1,4-dioxane solution via ultrasonication at 900 W for 30 min, ensuring a homogeneous suspension. The C-PANI/SBS/1,4-dioxane mixture was then transferred into a PTFE mold and frozen at −40°C overnight, followed by immediate freeze-drying to obtain the C-PANI/SBS foam. The C-PANI content in the C-PANI/SBS foam was set at 6 wt%. The other control samples of SBS foam were prepared using the same procedure as the C-PANI/SBS foam but without adding C-PANI.

Data preprocessing methods

The VS dataset comprises 6,124 paired samples, each containing an RGB image and signals from three tactile sensors (Closing eyes: 1,571 samples, Nodding: 1,341 samples, Normal: 1,822 samples, and Yawning: 1,390 samples). Data were collected from five volunteers (three males, two females) seated in the driver's seat of a stationary passenger vehicle. The camera-sensor setup was fixed across sessions, no auxiliary lighting was used, and images were acquired under two ambient-lighting regimes: (i) normal daylight and (ii) low-light conditions in an indoor parking garage. Accordingly, the visual background was the cabin interior (dashboard, steering wheel, seat), with natural variation in luminance and shadowing. Raw images were resized to 224 × 224 pixels for uniformity. The three tactile sensors were placed near the eyes, mouth, and neck to capture localized skin variations. To evaluate robustness, we additionally constructed six stress-test subsets spanning four appearance factors—older adults, darker skin tones, dense facial hair, and medical masks—and two challenging conditions: very low illumination with severe motion artifacts, and degraded sensor-skin adhesion. To mitigate inter-subject baseline differences introduced by the wearable device, the tactile-sensor capacitance change ($\Delta C/C_0$) was first low-pass filtered to suppress high-frequency noise and then processed with min-max normalization to ensure consistency across individuals. Informed written consent was obtained from all participants prior to their involvement in the study. All participants voluntarily took part in this study and provided written informed consent prior to enrollment. Because the dataset contains participants'

portrait images, all personally identifiable information was collected and used only after obtaining their explicit, fully informed written consent.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Code availability

The source code for the multimodal learning framework used in this study is available at <https://github.com/Createyou/Multimodal-Framework>

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Author contributions

K. L., W.Y. and Y. G. conceived the idea and planned the experiments. W.Y. and Y. G. performed the experiments and helped the realize of the sensor. K. L., D. Z. and K. B. designed the algorithm and developed the interface for demonstration. K. L., W.Y., D. S. and Y. G. contributed to the data analysis. K. L. and W.Y. wrote the paper. Y. L. and J. L. supervised the overall research. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available.

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References

- 1 Lal, S. K. & Craig, A. A critical review of the psychophysiology of driver fatigue. *Biol. Psychol.* **55**, 173-194 (2001).
- 2 Sikander, G. & Anwar, S. Driver Fatigue Detection Systems: A Review. *IEEE. Trans. Intell. Transp. Syst.* **20**, 2339-2352 (2019).

- 3 Li, Y. *et al.* A machine learning - assisted multifunctional tactile sensor for smart prosthetics. *InfoMat* **5** (2023).
- 4 Lu, Y., Liu, C., Chang, F., Liu, H. & Huan, H. JHPFA-Net: Joint Head Pose and Facial Action Network for Driver Yawning Detection Across Arbitrary Poses in Videos. *IEEE Trans. Intell. Transp. Syst.* **24**, 11850-11863 (2023).
- 5 Tang, Y. *et al.* Digital channel-enabled distributed force decoding via small datasets for hand-centric interactions. *Sci. Adv.* **11**, eadt2641 (2025).
- 6 Tashakori, A. *et al.* Capturing complex hand movements and object interactions using machine learning-powered stretchable smart textile gloves. *Nat. Mach. Intell.* **6**, 106-118 (2024).
- 7 Yang, L., Yang, H., Wei, H., Hu, Z. & Lv, C. Video-Based Driver Drowsiness Detection With Optimised Utilization of Key Facial Features. *IEEE Trans. Intell. Transp. Syst.* **25**, 6938-6950 (2024).
- 8 Huang, Y., Liu, C., Chang, F. & Lu, Y. Self-Supervised Multi-Granularity Graph Attention Network for Vision-Based Driver Fatigue Detection. *IEEE Trans. Emerging Top. Comput. Intell.* **8**, 3067-3080 (2024).
- 9 Li, R. *et al.* YOLO-SGC: A Dangerous Driving Behavior Detection Method With Multiscale Spatial-Channel Feature Aggregation. *IEEE Sens. J.* **24**, 36044-36056 (2024).
- 10 Liang, X., Yao, W., Fang, X. & Zhang, C. FMIF: facial multi-feature information fusion for driver fatigue detection. *Signal, Image Video Process.* **19**, 121 (2024).
- 11 Li, K., Gong, Y. & Ren, Z. A Fatigue Driving Detection Algorithm Based on Facial Multi-Feature Fusion. *IEEE Access* **8**, 101244-101259 (2020).
- 12 Murugan, S., Selvaraj, J. & Sahayadhas, A. Detection and analysis: driver state with electrocardiogram (ECG). *Phys. Eng. Sci. Med.* **43**, 525-537 (2020).
- 13 Chui, K. T., Tsang, K. F., Chi, H. R., Ling, B. W. K. & Wu, C. K. An Accurate ECG-Based Transportation Safety Drowsiness Detection Scheme. *IEEE Trans. Ind. Inf.* **12**, 1438-1452 (2016).
- 14 Jung, S. J., Shin, H. S. & Chung, W. Y. Driver fatigue and drowsiness monitoring system with embedded electrocardiogram sensor on steering wheel. *Iet. Intell. Transp. Sy.* **8**, 43-50 (2014).

- 15 Karthick, P. A., Ghosh, D. M. & Ramakrishnan, S. Surface electromyography based muscle fatigue detection using high-resolution time-frequency methods and machine learning algorithms. *Comput. Meth. Prog. Bio.* **154**, 45-56 (2018).
- 16 Zheng, W. L. *et al.* Vigilance Estimation Using a Wearable EOG Device in Real Driving Environment. *IEEE Trans. Intell. Transp. Syst.* **21**, 170-184 (2020).
- 17 Huang, Z. J. *et al.* In-Sensor Tactile Fusion and Logic for Accurate Intention Recognition. *Adv. Mater.* **36** (2024).
- 18 Cheng, A. B. *et al.* An intelligent hybrid-fabric wristband system enabled by thermal encapsulation for ergonomic human-machine interaction. *Nat. Commun.* **16** (2025).
- 19 Oh, S. *et al.* Easy-To-Wear Auxetic SMA Knot-Architecture for Spatiotemporal and Multimodal Haptic Feedbacks. *Adv. Mater.* **35**, e2304442 (2023).
- 20 Xu, Y. H. *et al.* Real-Time Monitoring System of Automobile Driver Status and Intelligent Fatigue Warning Based on Triboelectric Nanogenerator. *Acs Nano* **15**, 7271-7278 (2021).
- 21 Luo, F. Y. *et al.* Wearable and self-powered triboelectric sensors based on NaCl/PVA hydrogel for driver multidimensional information monitoring. *Nano Energy* **118** (2023).
- 22 Lu, X. *et al.* Stretchable, transparent triboelectric nanogenerator as a highly sensitive self-powered sensor for driver fatigue and distraction monitoring. *Nano Energy* **78** (2020).
- 23 Huang, J. *et al.* Recent advances and future prospects in tactile sensors for normal and shear force detection, decoupling, and applications. *J. Semicond.* **45**, 121601 (2024).
- 24 Lee, H. *et al.* Vialess heterogeneous skin patch for multimodal monitoring and stimulation. *Nat. Commun.* **16** (2025).
- 25 Du, G. L. *et al.* A Multimodal Fusion Fatigue Driving Detection Method Based on Heart Rate and PERCLOS. *IEEE Trans. Intell. Transp. Syst.* **23**, 21810-21820 (2022).
- 26 Chen, Z., Hua, Q. & Shen, G. Flexible ultrasound arrays with embossed polymer structures for medical imaging. *J. Semicond.* **45**, 080401 (2024).
- 27 Liu, M. W. *et al.* A star-nose-like tactile-olfactory bionic sensing array for robust object recognition in non-visual environments. *Nat. Commun.* **13** (2022).
- 28 Wang, M. *et al.* Gesture recognition using a bioinspired learning architecture that integrates visual data with somatosensory data from stretchable sensors. *Nat. Electron.* **3**, (9):563-570 (2020).

- 29 Mao, Q., Liao, Z. J., Yuan, J. F. & Zhu, R. Multimodal tactile sensing fused with vision for dexterous robotic housekeeping. *Nat. Commun.* **15** (2024).
- 30 Babadian, R. P., Faez, K., Amiri, M. & Falotico, E. Fusion of tactile and visual information in deep learning models for object recognition. *Inform Fusion* **92**, 313-325 (2023).
- 31 Liu, Y., Wang, K. Z., Li, G. B. & Lin, L. Semantics-Aware Adaptive Knowledge Distillation for Sensor-to-Vision Action Recognition. *IEEE Trans. Image Process.* **30**, 5573-5588 (2021).
- 32 Sigrist, R., Rauter, G., Riener, R. & Wolf, P. Augmented visual, auditory, haptic, and multimodal feedback in motor learning: a review. *Psychon. Bull. Rev.* **20**, 21-53 (2013).
- 33 Ha, K. H. *et al.* Highly Sensitive Capacitive Pressure Sensors over a Wide Pressure Range Enabled by the Hybrid Responses of a Highly Porous Nanocomposite. *Adv. Mater.* **33**, e2103320 (2021).
- 34 Wang, P. *et al.* Tailorable Capacitive Tactile Sensor Based on Stretchable and Dissolvable Porous Silver Nanowire/Polyvinyl Alcohol Nanocomposite Hydrogel for Wearable Human Motion Detection. *Adv. Mater. Interfaces* **8** (2021).
- 35 Chhetry, A., Sharma, S., Yoon, H., Ko, S. & Park, J. Y. Enhanced sensitivity of capacitive pressure and strain sensor based on CaCu₃Ti₄O₁₂ wrapped hybrid sponge for wearable applications. *Adv. Funct. Mater.* **30** (2020).
- 36 Lü, X. Z., Yu, T. T., Meng, F. C. & Bao, W. M. Wide-Range and High-Stability Flexible Conductive Graphene/Thermoplastic Polyurethane Foam for Piezoresistive Sensor Applications. *Adv Mater Technol-Us* **6** (2021).
- 37 Zhong, D. L. *et al.* High-speed and large-scale intrinsically stretchable integrated circuits. *Nature* **627** (2024).
- 38 Nan, Z., Wei, W., Lin, Z. H., Chang, J. J. & Hao, Y. Flexible Nanocomposite Conductors for Electromagnetic Interference Shielding. *Nano-Micro Lett.* **15** (2023).
- 39 Tian, D. K. *et al.* In-situ metallized carbon nanotubes/poly(styrene-butadiene-styrene) (CNTs/SBS) foam for electromagnetic interference shielding. *Chem Eng J* **420** (2021).
- 40 Chu, X., Yang, W. Q. & Li, H. Recent advances in polyaniline-based micro-supercapacitors. *Mater. Horiz.* **10**, 670-697 (2023).

- 41 Zhang, L. Y. *et al.* Polytoluidine blue-coated multiwalled carbon nanotubes: In-situ polymerization based hybrid nanostructure for high performance asymmetric supercapacitors. *Mater. Today Chem.* **39** (2024).
- 42 Hao, S. W., Fu, Q. J., Meng, L., Xu, F. & Yang, J. A biomimetic laminated strategy enabled strain-interference free and durable flexible thermistor electronics. *Nat. Commun.* **13** (2022).
- 43 Yang, L. F. *et al.* Interior design of hierarchical micro/nanostructures for enhancing energy storage ability of polyanilines through frozen interfacial polymerization. *Electrochim Acta* **386** (2021).
- 44 Yu, C. J. *et al.* Chronological adhesive cardiac patch for synchronous mechanophysiological monitoring and electrocoupling therapy. *Nat. Commun.* **14** (2023).
- 45 Tan, M. X. & Le, Q. V. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *Pr. Mach. Learn. Res.* **97** (2019).
- 46 Krizhevsky, A., Sutskever, I. & Hinton, G. E. ImageNet Classification with Deep Convolutional Neural Networks. *Commun. Acm.* **60**, 84-90 (2017).
- 47 Wu, R. H. *et al.* Stretchable spring-sheathed yarn sensor for 3D dynamic body reconstruction assisted by transfer learning. *Infomat* **6**, (2024).
- 48 Kim, K. K. *et al.* A substrate-less nanomesh receptor with meta-learning for rapid hand task recognition. *Nat. Electron.* **6**, 64-75 (2022).
- 49 Van der Maaten, L. & Hinton, G. Visualizing Data using t-SNE. *J. Mach. Learn. Res.* **9**, 2579-2605 (2008).
- 50 Selvaraju, R. R. *et al.* Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *Int. J. Comput. Vision* **128**, 336-359 (2020).
- 51 Vaswani, A. *et al.* Attention is all you need. *Advances in Neural Information Processing Systems* (NIPS), (2017).
- 52 Dosovitskiy, A. *et al.* An image is worth 16x16 words: Transformers for image recognition at scale. *In Proc. 9th International Conference on Learning Representations* (2021).
- 53 Liu, Z. *et al.* Swin transformer: Hierarchical vision transformer using shifted windows. *In Proceedings of the IEEE/CVF international conference on computer vision* 10012–10022 (2021).
- 54 Liu, Z. *et al.* A convnet for the 2020s. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* 11976–11986 (2022).

- 55 Du, G., Li, T., Li, C., Liu, P. X. & Li, D. Vision-Based Fatigue Driving Recognition Method Integrating Heart Rate and Facial Features. *IEEE Trans. Intell. Transp. Syst.* **22**, 3089–3100 (2021).
- 56 Shi, J. & Wang, K. Fatigue driving detection method based on Time-Space-Frequency features of multimodal signals. *Biomed. Signal Process. Control* **84**, 104744 (2023).
- 57 Hassan, O. F., Ibrahim, A. F., Gomaa, A., Makhoul, M. A. & Hafiz, B. Real-time driver drowsiness detection using transformer architectures: a novel deep learning approach. *Sci. Rep.* **15**, 17493 (2025).
- 58 Xing, Y. et al. MS-DETR: Multispectral Pedestrian Detection Transformer With Loosely Coupled Fusion and Modality-Balanced Optimization. *IEEE Trans. Intell. Transport. Syst.* **25**, 20628–20642 (2024).
- 59 Li, Z., Zhao, X., Wu, F., Chen, D. & Wang, C. A Lightweight and Efficient Distracted Driver Detection Model Fusing Convolutional Neural Network and Vision Transformer. *IEEE Trans. Intell. Transport. Syst.* **25**, 19962–19978 (2024).

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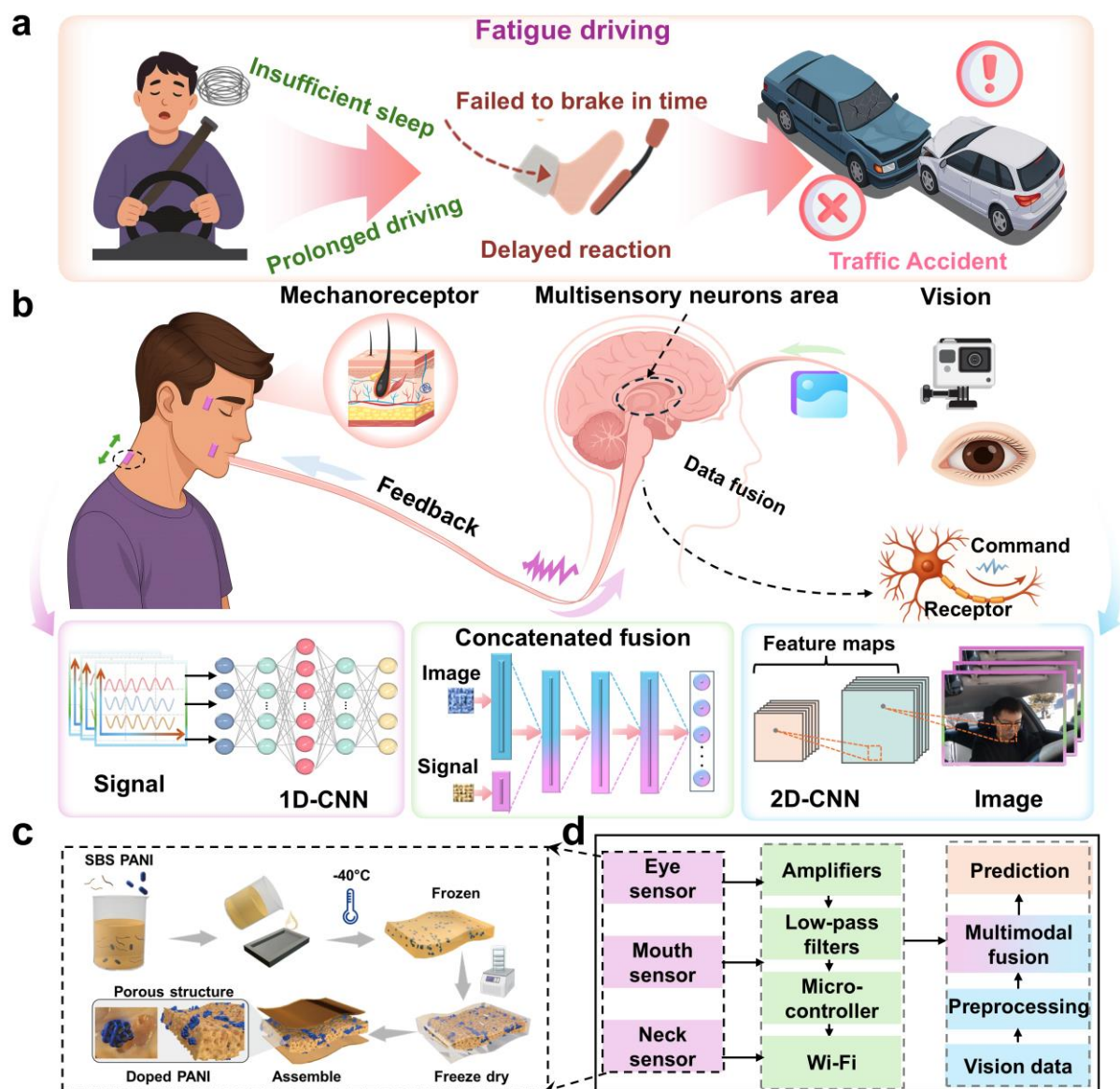


Fig. 1 Illustration of multimodal feature fusion for fatigue driving detection. **(a)** Car accidents caused by driver fatigue. **(b)** Schematic of vision and tactile information processing in the human brain and framework for sensor-vision data fusion using CNN. **(c)** Schematic illustration of the fabrication of flexible sensors composed of porous composites. **(d)** System-level diagram of multimodal driver fatigue monitoring.

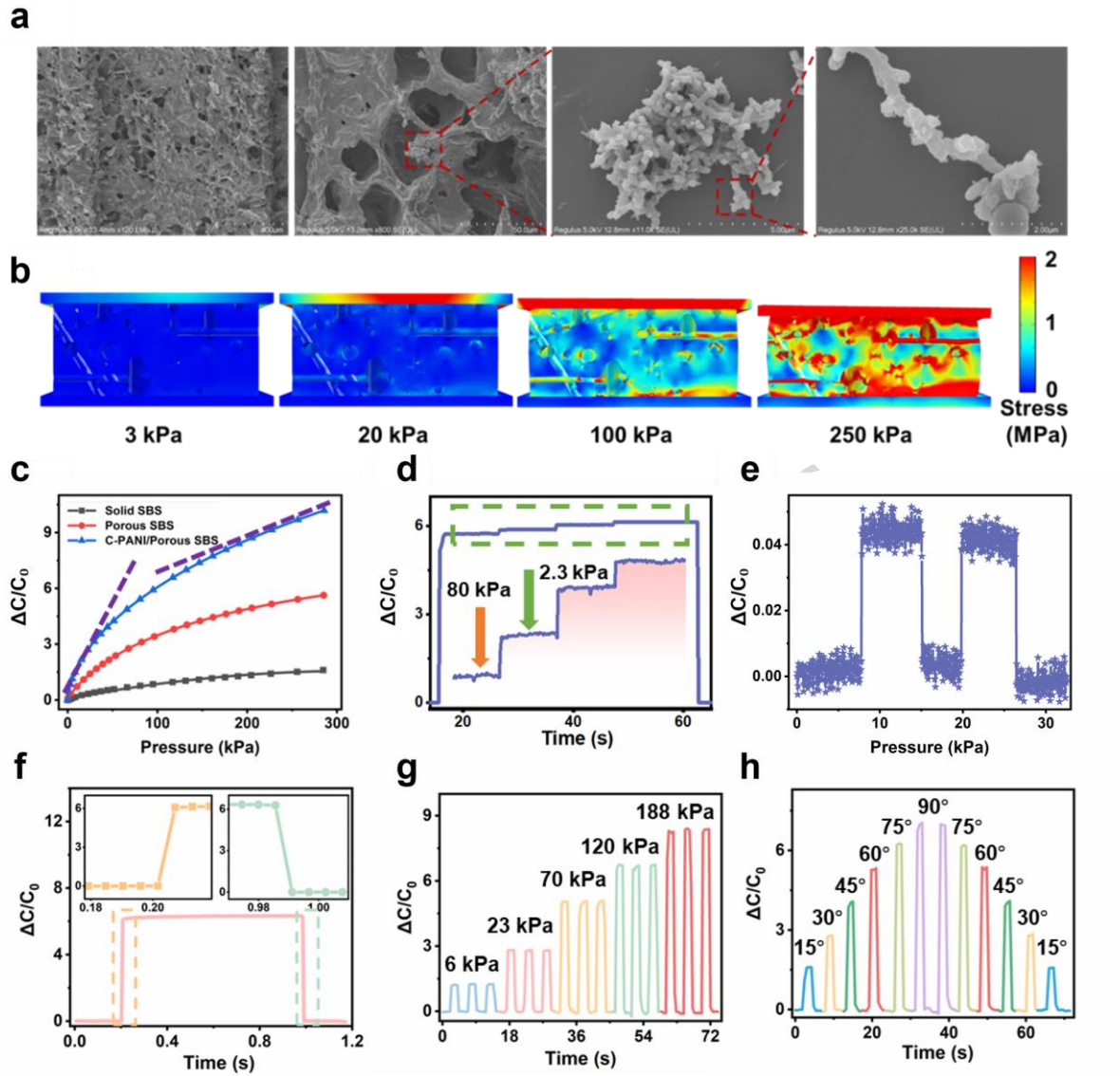


Fig. 2 (a) SEM images of the dielectric layer characterizing micron-scale pores and nanoscale C-PANI at different magnifications. (b) Finite element simulation results of the cross-sectional stress distribution in the SBS foam structure model under different pressures. (c) Relative capacitance changes of three capacitive tactile sensors: solid SBS layer, pure SBS foam, and SBS foam with C-PANI. (d) Relative capacitance changes of the sensor for slight pressure variations under high pressure conditions. (e) Consecutive capacitive responses at the limit of detection. (f) Response and recovery times of the sensor. (g) Capacitive response of the sensor to repeatedly applied gradient pressure. (h) Capacitive response of the sensor to continuous gradient increases and decreases in the bending angle.

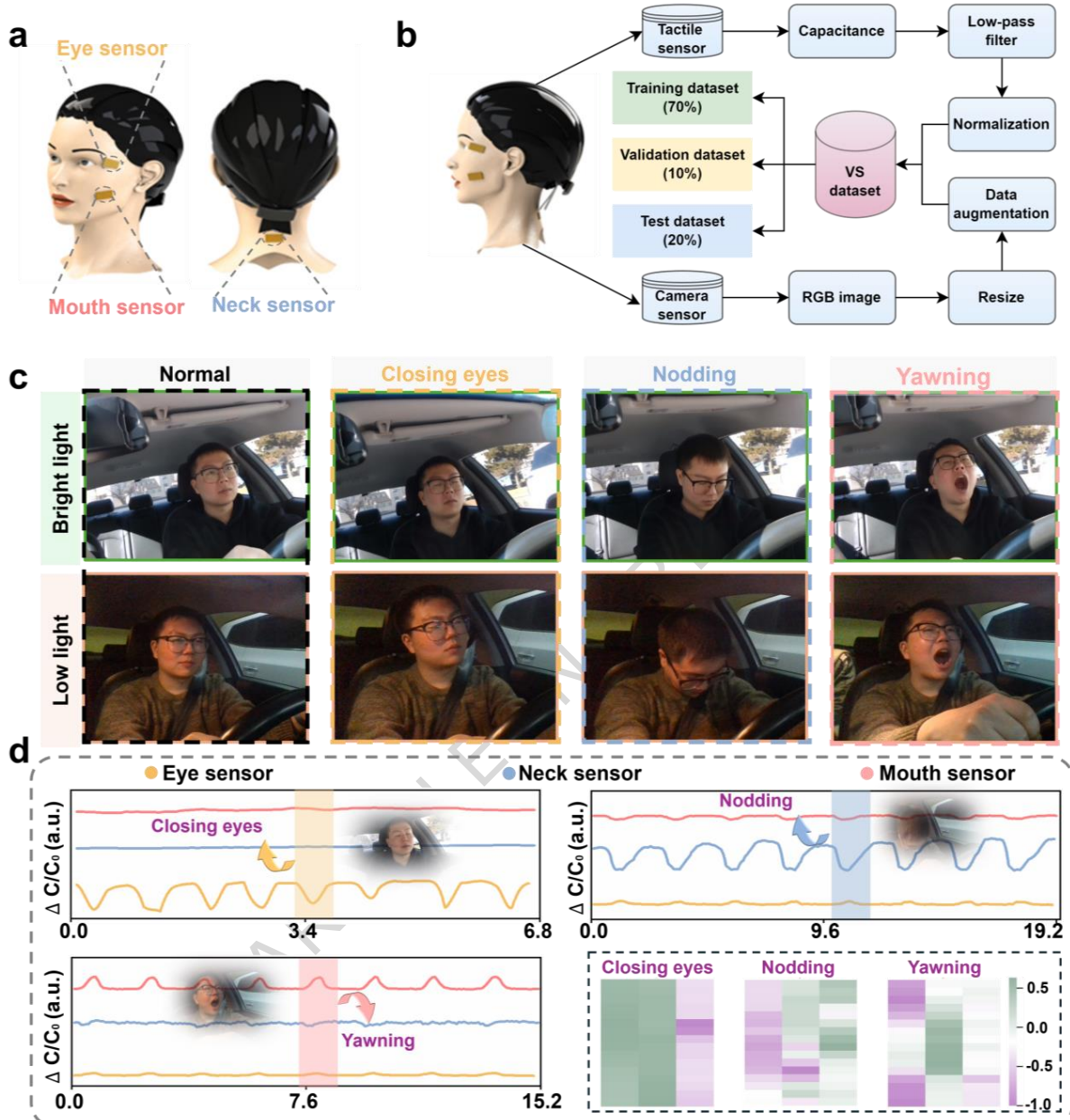


Fig. 3 Data preparation for fatigue driving detection based on vision and sensor fusion. **(a)** Placement illustration of a flexible sensor on the mouth, eye, and neck. **(b)** Flow diagram for data collection to construct the VS dataset from tactile and camera sensors. **(c)** Illustration of four driving states under different illuminance conditions with vision images. **(d)** Real-time three-channel waveforms and corresponding heatmaps for three fatigue states.

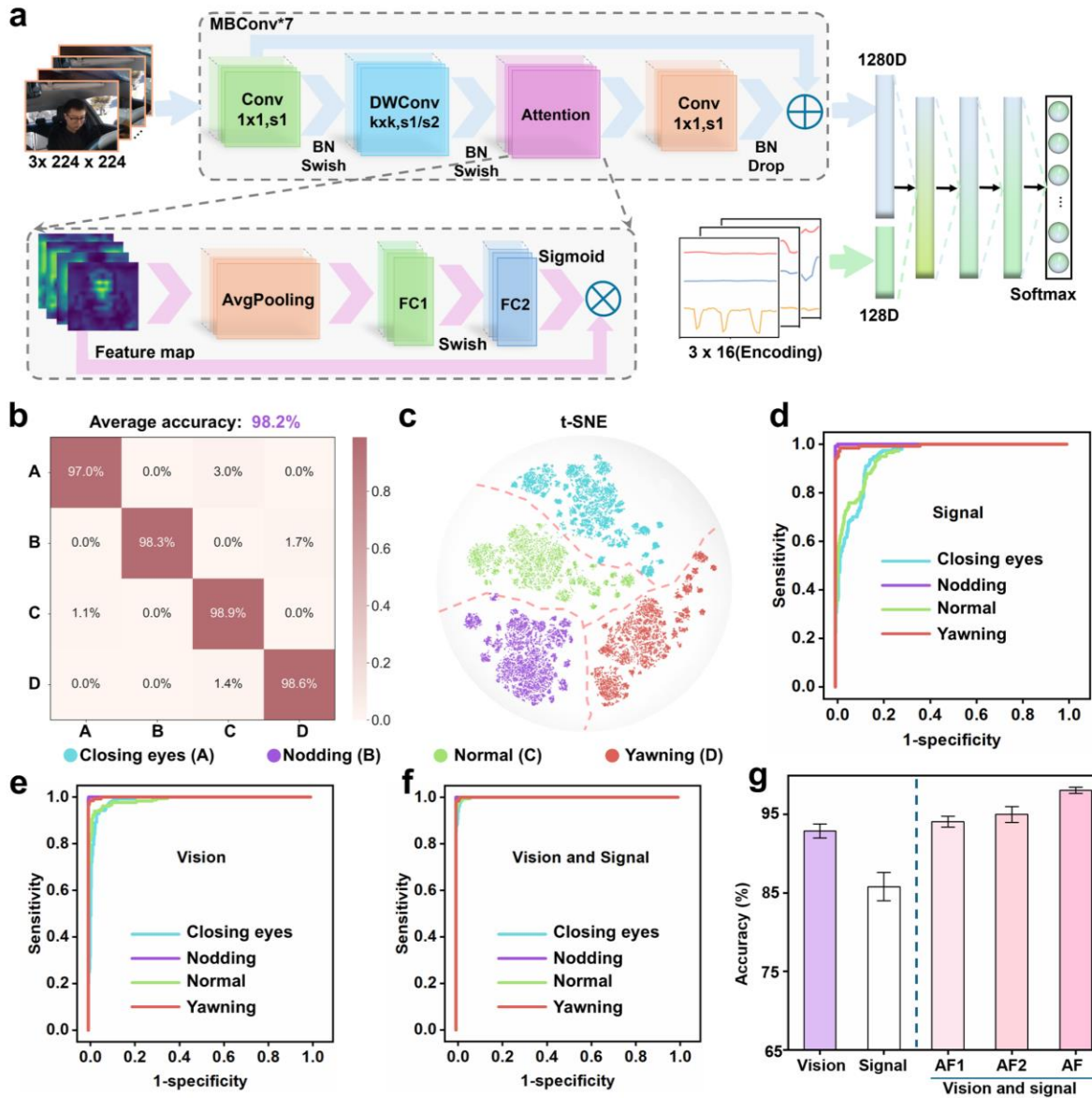


Fig. 4 Multimodal model for fatigue driving detection. (a) Detailed architecture of the multimodal model. (b) Confusion matrix for predicting four driving states with an average accuracy of 98.2%. (c) t-SNE visualization of feature distribution for the four driving states. (d) ROC curve of sensor-based fatigue detection. (e) ROC curve of vision-based fatigue detection. (f) ROC curve of VS fatigue detection. (g) Vision-based recognition uses only images ($92.9 \pm 0.7\%$), and signal-based recognition uses only tactile sensor data ($85.8 \pm 1.8\%$). AF1: weighted averaging fusion ($94.1 \pm 0.7\%$), AF2: weighted-multiplication fusion ($95.0 \pm 1.0\%$). AF: concatenation fusion ($97.9 \pm 0.4\%$).

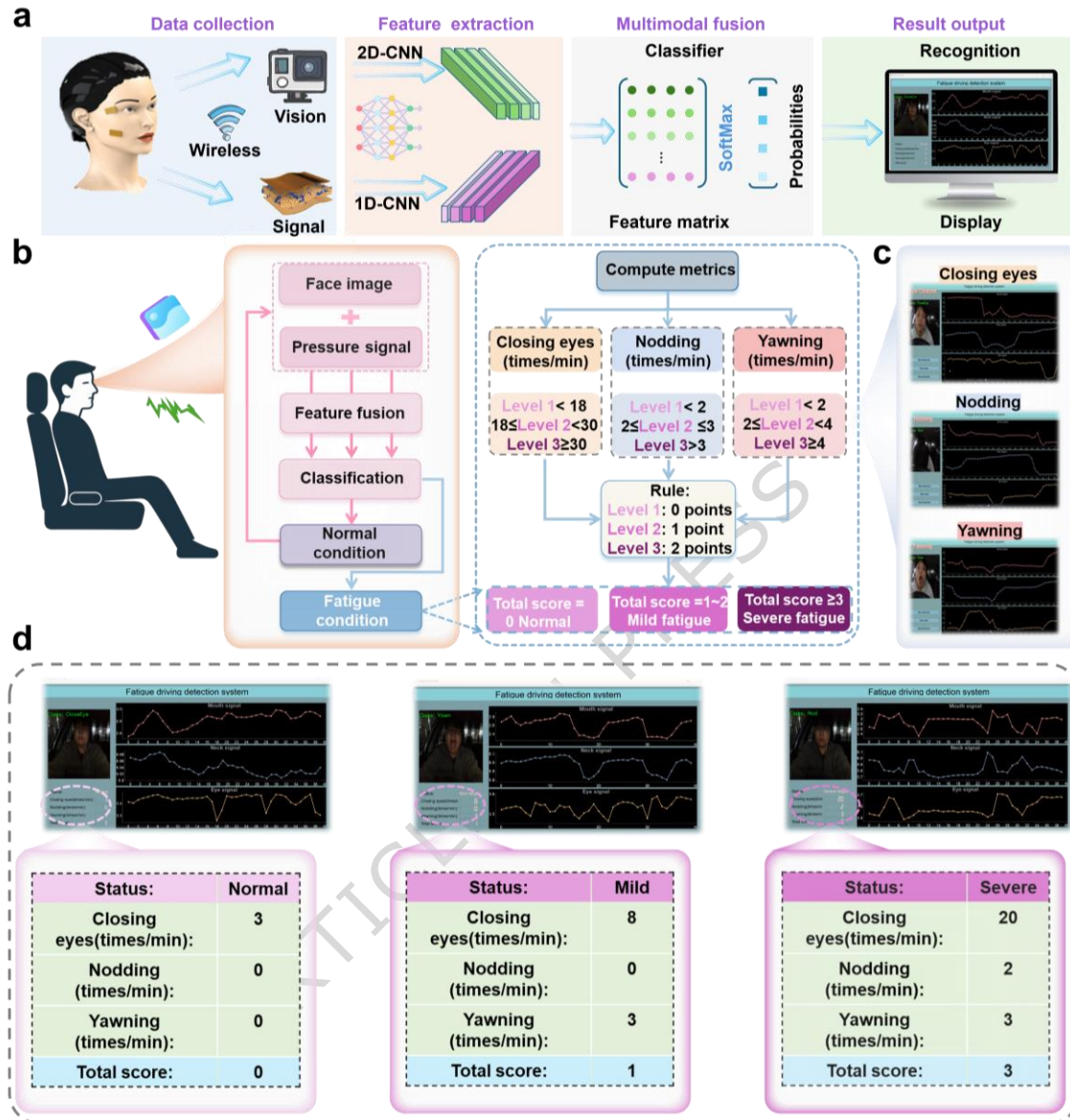


Fig. 5 Demonstration of fatigue driving detection in actual low-light environments. (a) Illustration of the process of multimodal data pre-processing, feature fusion, and prediction for fatigue driving monitoring. **(b)** Decision framework for multimodal fatigue monitoring based on vision and tactile signals. **(c)** Real-time prediction of fatigue status. **(d)** Detailed evaluation of fatigue status.